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Roll Number : A20 Section : Q5E05	Academic Task Type : Individual
Registration Number : 12521887	Course Code : INTM 590
Course Title : Big Data Analytics	Submitted To : Rinki Gupta Mam
Date of Allotment : 24/04/26	Date of Submission : 30/04/26

Learning Outcomes:

This Big Data Analytics Project significantly enhanced my understanding of Big Data concepts and practical implementation using Apache Sqoop, Apache Pig, and Apache Hive within the Hadoop ecosystem. Through the completion of various practical tasks, I gained hands-on experience in data import/export operations, distributed data processing, and data warehousing techniques in big data environments.

During Task 1, I learned how to use Apache Sqoop for efficient data transfer between MySQL databases and Hadoop Distributed File System (HDFS). I gained practical knowledge of creating databases and tables in MySQL, inserting records, and using Sqoop commands such as import, import-all-tables, and export. I understood how relational data can be imported into HDFS and exported back to databases while maintaining data consistency. The use of parameters like --connect, --username, --password, --target-dir, --warehouse-dir, and -m helped me understand how Sqoop integrates traditional databases with distributed storage systems.

I also developed a better understanding of how MapReduce jobs are automatically executed during Sqoop operations and how data is stored in HDFS using partitioned output files such as part-m-00000 along with _SUCCESS indicators. This practical exposure improved my understanding of large-scale data migration and management in Hadoop environments.

Furthermore, Task 2 provided me with practical exposure to Apache Pig and Pig Latin scripting for data analysis and transformation. I learned how to load structured data from HDFS using the LOAD command with PigStorage, define schemas, and process records using operations such as FOREACH, GROUP BY, ORDER BY, and FILTER. These operations helped me understand how data can be transformed, grouped, sorted, and filtered efficiently in distributed systems.

Through Pig analytics commands, I gained valuable experience in handling semi-structured datasets and performing analytical operations similar to SQL queries but optimized for Hadoop processing. I also understood how Pig internally converts scripts into MapReduce jobs for distributed execution. This improved my confidence in performing big data analytics tasks using scripting-based tools.

In addition, Task 3 enhanced my knowledge of Apache Hive and data warehousing concepts in Hadoop. I learned how to create and manage internal tables, external tables, static partitions, and dynamic partitions. I understood the difference between managed and external tables and how Hive stores and accesses data in HDFS. I also learned how partitioning improves query performance and data organization in large datasets.

The implementation of static and dynamic partitioning gave me practical insight into how Hive optimizes data storage and retrieval in real-world big data applications. I also became familiar with Hive Query Language (HQL) commands used for creating tables, loading data, inserting partitions, and querying datasets.

Overall, this project strengthened both my theoretical and practical understanding of Big Data Analytics. It improved my technical skills in using Hadoop ecosystem tools such as Sqoop, Pig, Hive, HDFS, and MapReduce. The project also enhanced my ability to process, transfer, store, and analyze large datasets efficiently in distributed computing environments, which are essential skills in modern data-driven industries.

Declaration:

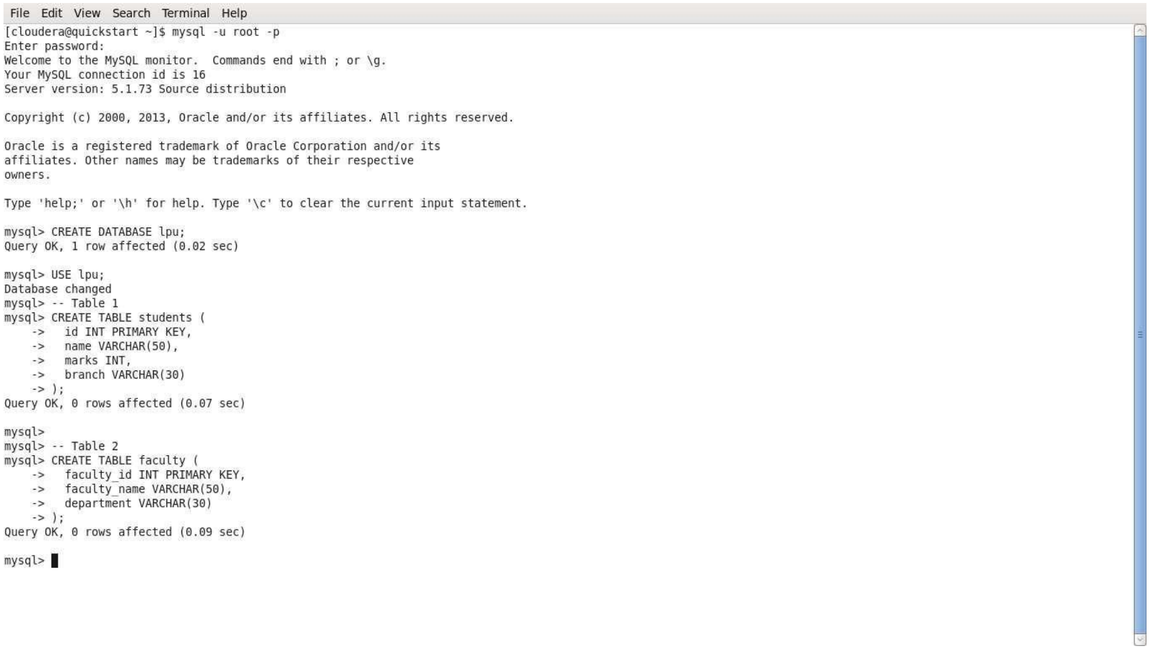
I hereby declare that this project is entirely my own work and has been completed independently. I confirm that I have not copied or reproduced any part of this work from other students, online sources, or any external material without proper acknowledgment.

I further affirm that no part of this project has been written, prepared, or submitted by any other individual on my behalf. This submission reflects my own understanding, efforts, and learning derived from the course material and practical exercises. I take full responsibility for the originality and authenticity of the content presented in this project.

TASK 1 — SQOOP: Data Import and Export

Database Setup in MySQL

First we logged into MySQL and created a database called lpu. Then we made three tables: students, faculty, and students_export. The students table has id, name, marks, and branch columns.



```
File  Edit  View  Search  Terminal  Help
[cloudera@quickstart ~]$ mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 16
Server version: 5.1.73 Source distribution

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE lpu;
Query OK, 1 row affected (0.02 sec)

mysql> USE lpu;
Database changed
mysql> -- Table 1
mysql> CREATE TABLE students (
    ->   id INT PRIMARY KEY,
    ->   name VARCHAR(50),
    ->   marks INT,
    ->   branch VARCHAR(30)
    -> );
Query OK, 0 rows affected (0.07 sec)

mysql>
mysql> -- Table 2
mysql> CREATE TABLE faculty (
    ->   faculty_id INT PRIMARY KEY,
    ->   faculty_name VARCHAR(50),
    ->   department VARCHAR(30)
    -> );
Query OK, 0 rows affected (0.09 sec)

mysql>
```
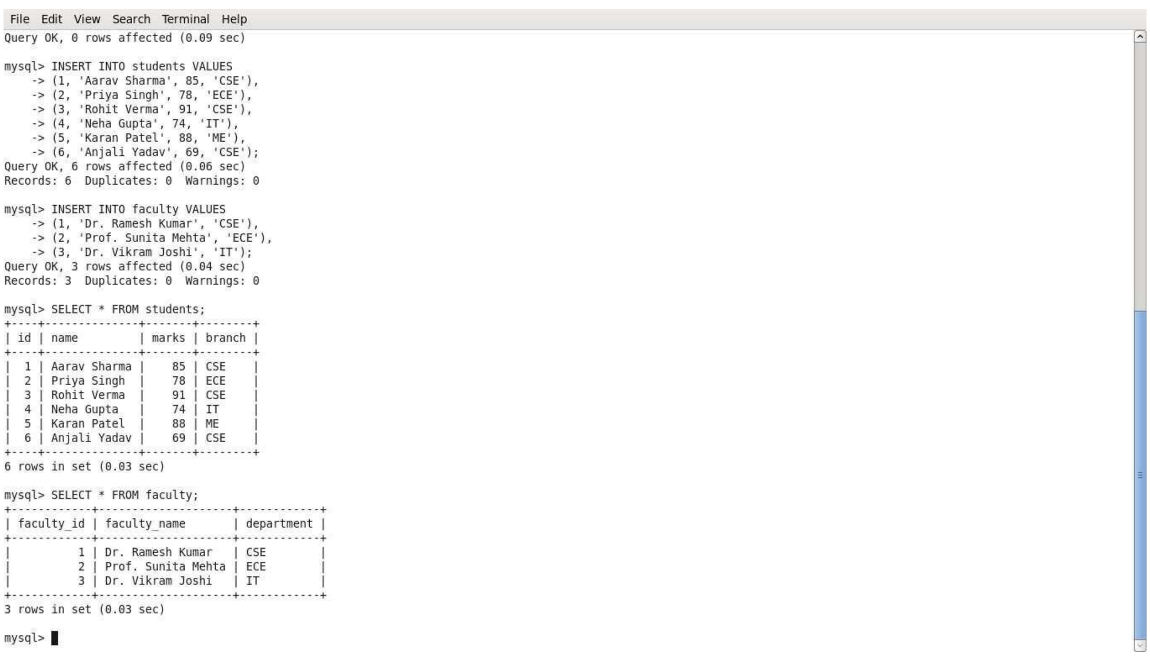

Screenshot 1: MySQL login, database lpu created, and tables students and faculty created.

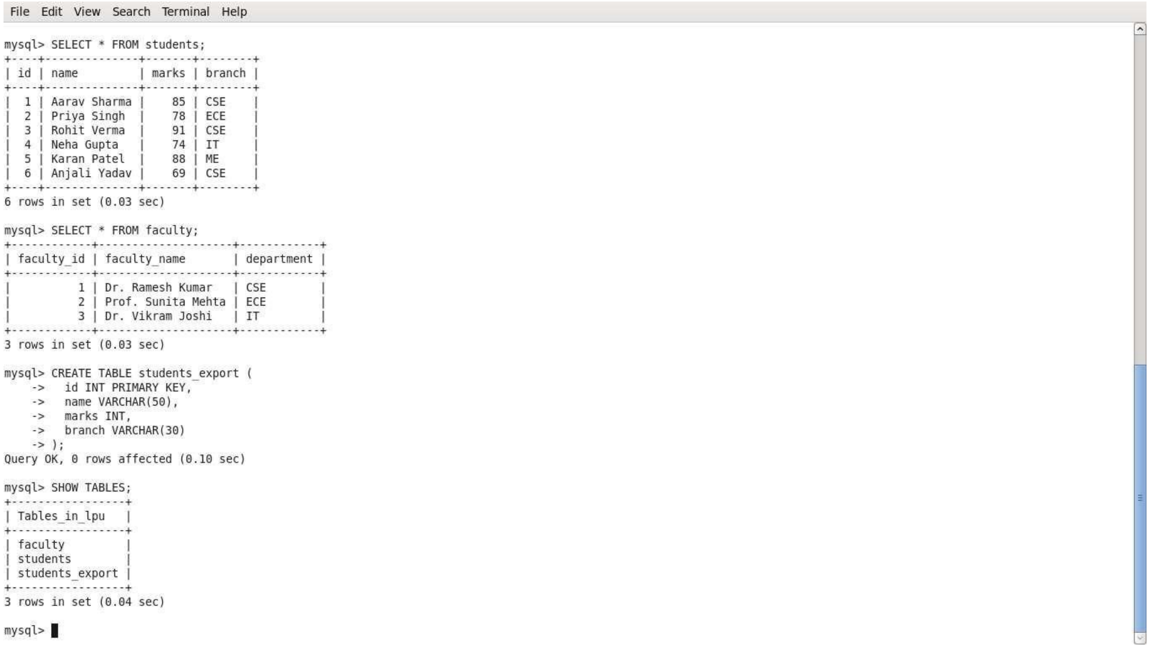
Next we inserted 6 student records and 3 faculty records, then ran SELECT * to confirm the data is there.



```
File  Edit  View  Search  Terminal  Help
Query OK, 0 rows affected (0.09 sec)

mysql> INSERT INTO students VALUES
    -> (1, 'Aarav Sharma', 85, 'CSE'),
    -> (2, 'Priya Singh', 78, 'ECE'),
    -> (3, 'Rohit Verma', 91, 'CSE'),
    -> (4, 'Neha Gupta', 74, 'IT'),
    -> (5, 'Karan Patel', 88, 'ME'),
    -> (6, 'Anjali Yadav', 69, 'CSE');
Query OK, 6 rows affected (0.06 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql> INSERT INTO faculty VALUES
    -> (1, 'Dr. Ramesh Kumar', 'CSE'),
    -> (2, 'Prof. Sunita Mehta', 'ECE'),
    -> (3, 'Dr. Vikram Joshi', 'IT');
Query OK, 3 rows affected (0.04 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM students;
+----+--------------+-------+--------+
| id | name         | marks | branch |
+----+--------------+-------+--------+
|  1 | Aarav Sharma |    85 | CSE    |
|  2 | Priya Singh  |    78 | ECE    |
|  3 | Rohit Verma  |    91 | CSE    |
|  4 | Neha Gupta   |    74 | IT     |
|  5 | Karan Patel  |    88 | ME     |
|  6 | Anjali Yadav |    69 | CSE    |
+----+--------------+-------+--------+
6 rows in set (0.03 sec)

mysql> SELECT * FROM faculty;
+------------+--------------------+------------+
| faculty_id | faculty_name       | department |
+------------+--------------------+------------+
|          1 | Dr. Ramesh Kumar   | CSE        |
|          2 | Prof. Sunita Mehta | ECE        |
|          3 | Dr. Vikram Joshi   | IT         |
+------------+--------------------+------------+
3 rows in set (0.03 sec)

mysql>
```

Screenshot 2: Data inserted into students and faculty tables. SELECT confirms 6 students and 3 faculty rows.

We also created the students_export table (same structure as students) and ran SHOW TABLES to confirm all 3 tables exist in lpu.



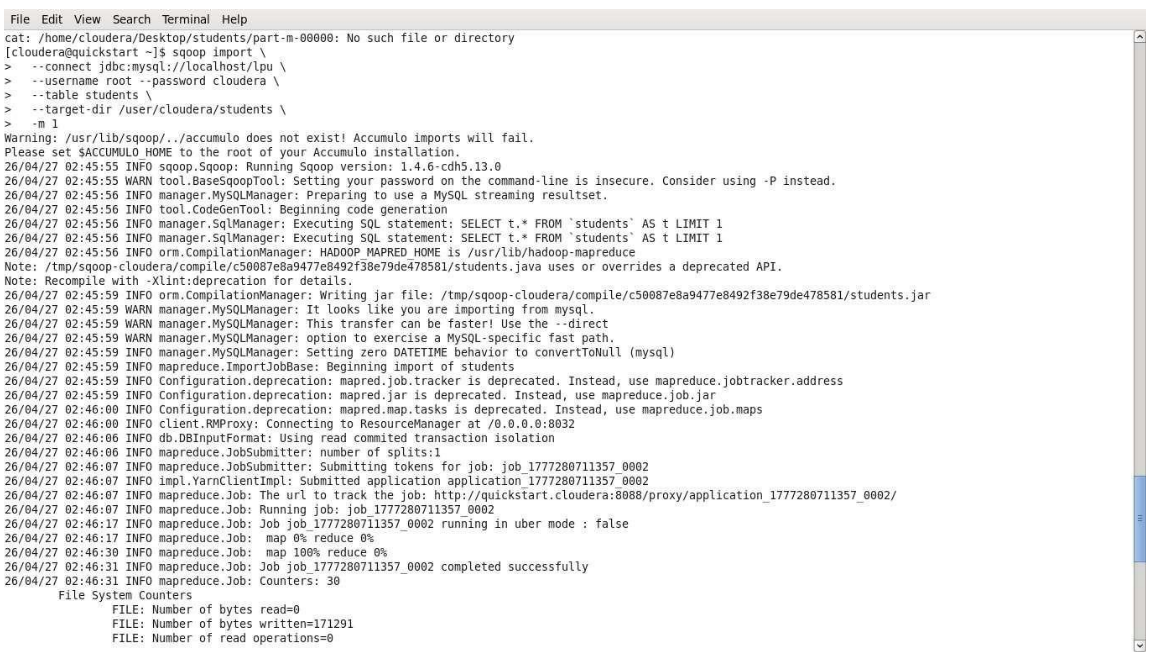
```
File  Edit  View  Search  Terminal  Help

mysql> SELECT * FROM students;
+----+--------------+-------+--------+
| id | name         | marks | branch |
+----+--------------+-------+--------+
|  1 | Aarav Sharma |    85 | CSE    |
|  2 | Priya Singh  |    78 | ECE    |
|  3 | Rohit Verma  |    91 | CSE    |
|  4 | Neha Gupta   |    74 | IT     |
|  5 | Karan Patel  |    88 | ME     |
|  6 | Anjali Yadav |    69 | CSE    |
+----+--------------+-------+--------+
6 rows in set (0.03 sec)

mysql> SELECT * FROM faculty;
+------------+-------------------+------------+
| faculty_id | faculty_name      | department |
+------------+-------------------+------------+
|          1 | Dr. Ramesh Kumar  | CSE        |
|          2 | Prof. Sunita Mehta | ECE       |
|          3 | Dr. Vikram Joshi  | IT         |
+------------+-------------------+------------+
3 rows in set (0.03 sec)

mysql> CREATE TABLE students_export (
    ->   id INT PRIMARY KEY,
    ->   name VARCHAR(50),
    ->   marks INT,
    ->   branch VARCHAR(30)
    -> );
Query OK, 0 rows affected (0.10 sec)

mysql> SHOW TABLES;
+-----------------+
| Tables_in_lpu   |
+-----------------+
| faculty         |
| students        |
| students_export |
+-----------------+
3 rows in set (0.04 sec)

mysql>
```

Screenshot 3: students_export table created. SHOW TABLES shows all 3 tables: faculty, students, students_export.

Q1: Import a Single Table from MySQL to HDFS

Command used: sqoop import with --connect, --username, --password, --table students, --target-dir /user/cloudera/students, -m 1

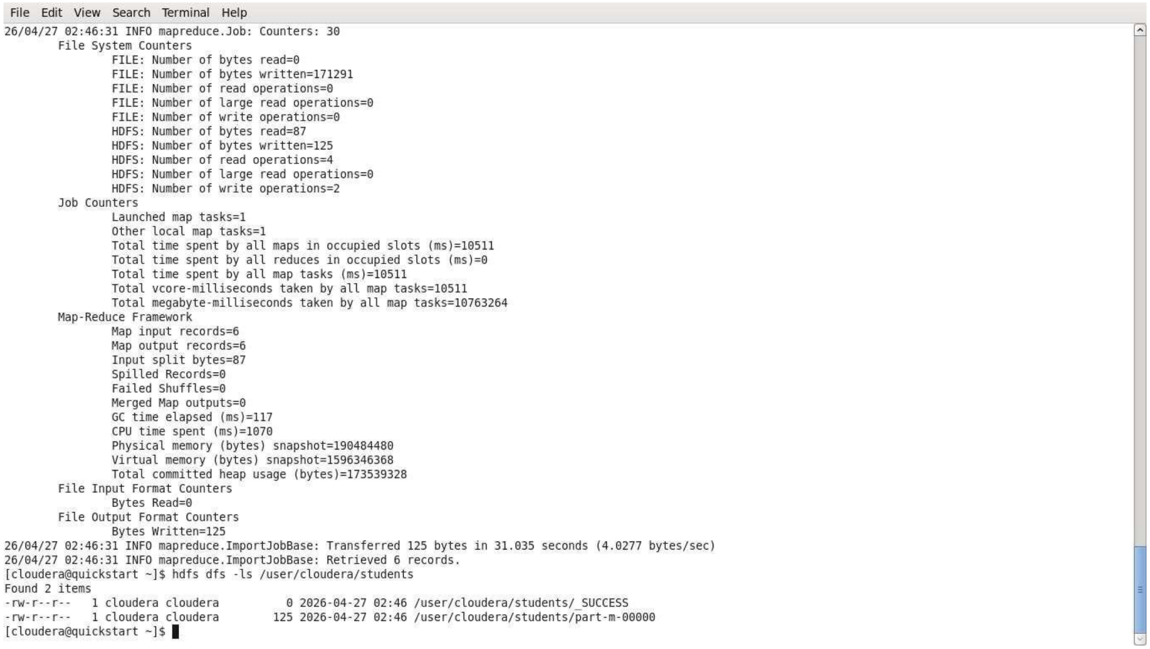
This pulls the students table from MySQL into HDFS at the path /user/cloudera/students. The -m 1 flag means we use only 1 mapper.



```
File  Edit  View  Search  Terminal  Help
cat: /home/cloudera/Desktop/students/part-m-00000: No such file or directory
[cloudera@quickstart ~]$ sqoop import \
>   --connect jdbc:mysql://localhost/lpu \
>   --username root --password cloudera \
>   --table students \
>   --target-dir /user/cloudera/students \
>   -m 1
Warning: /usr/lib/sqoop/../accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
26/04/27 02:45:55 INFO sqoop.Sqoop: Running Sqoop version: 1.4.6-cdh5.13.0
26/04/27 02:45:55 WARN tool.BaseSqoopTool: Setting your password on the command-line is insecure. Consider using -P instead.
26/04/27 02:45:56 INFO manager.MySQLManager: Preparing to use a MySQL streaming resultset.
26/04/27 02:45:56 INFO tool.CodeGenTool: Beginning code generation
26/04/27 02:45:56 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM `students` AS t LIMIT 1
26/04/27 02:45:56 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM `students` AS t LIMIT 1
26/04/27 02:45:56 INFO orm.CompilationManager: HADOOP_MAPRED_HOME is /usr/lib/hadoop-mapreduce
Note: /tmp/sqoop-cloudera/compile/c50087e8a9477e8492f38e79de478581/students.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
26/04/27 02:45:59 INFO orm.CompilationManager: Writing jar file: /tmp/sqoop-cloudera/compile/c50087e8a9477e8492f38e79de478581/students.jar
26/04/27 02:45:59 WARN manager.MySQLManager: It looks like you are importing from mysql.
26/04/27 02:45:59 WARN manager.MySQLManager: This transfer can be faster! Use the --direct
26/04/27 02:45:59 WARN manager.MySQLManager: option to exercise a MySQL-specific fast path.
26/04/27 02:45:59 INFO manager.MySQLManager: Setting zero DATETIME behavior to convertToNull (mysql)
26/04/27 02:45:59 INFO mapreduce.ImportJobBase: Beginning import of students
26/04/27 02:45:59 INFO Configuration.deprecation: mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
26/04/27 02:45:59 INFO Configuration.deprecation: mapred.jar is deprecated. Instead, use mapreduce.job.jar
26/04/27 02:46:00 INFO Configuration.deprecation: mapred.map.tasks is deprecated. Instead, use mapreduce.job.maps
26/04/27 02:46:00 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
26/04/27 02:46:06 INFO db.DBInputFormat: Using read commited transaction isolation
26/04/27 02:46:06 INFO mapreduce.JobSubmitter: number of splits:1
26/04/27 02:46:07 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1777280711357_0002
26/04/27 02:46:07 INFO impl.YarnClientImpl: Submitted application application_1777280711357_0002
26/04/27 02:46:07 INFO mapreduce.Job: The url to track the job: http://quickstart.cloudera:8088/proxy/application_1777280711357_0002/
26/04/27 02:46:07 INFO mapreduce.Job: Running job: job_1777280711357_0002
26/04/27 02:46:17 INFO mapreduce.Job: Job job_1777280711357_0002 running in uber mode : false
26/04/27 02:46:17 INFO mapreduce.Job:  map 0% reduce 0%
26/04/27 02:46:30 INFO mapreduce.Job:  map 100% reduce 0%
26/04/27 02:46:31 INFO mapreduce.Job: Job job_1777280711357_0002 completed successfully
26/04/27 02:46:31 INFO mapreduce.Job: Counters: 30
        File System Counters
                FILE: Number of bytes read=0
                FILE: Number of bytes written=171291
                FILE: Number of read operations=0
```

Screenshot 4: Sqoop import command running. MapReduce job starts — map 0% then 100%.

After the job finishes, we checked HDFS with hdfs dfs -ls. We can see a _SUCCESS file and part-m-00000 (the actual data file, 125 bytes).



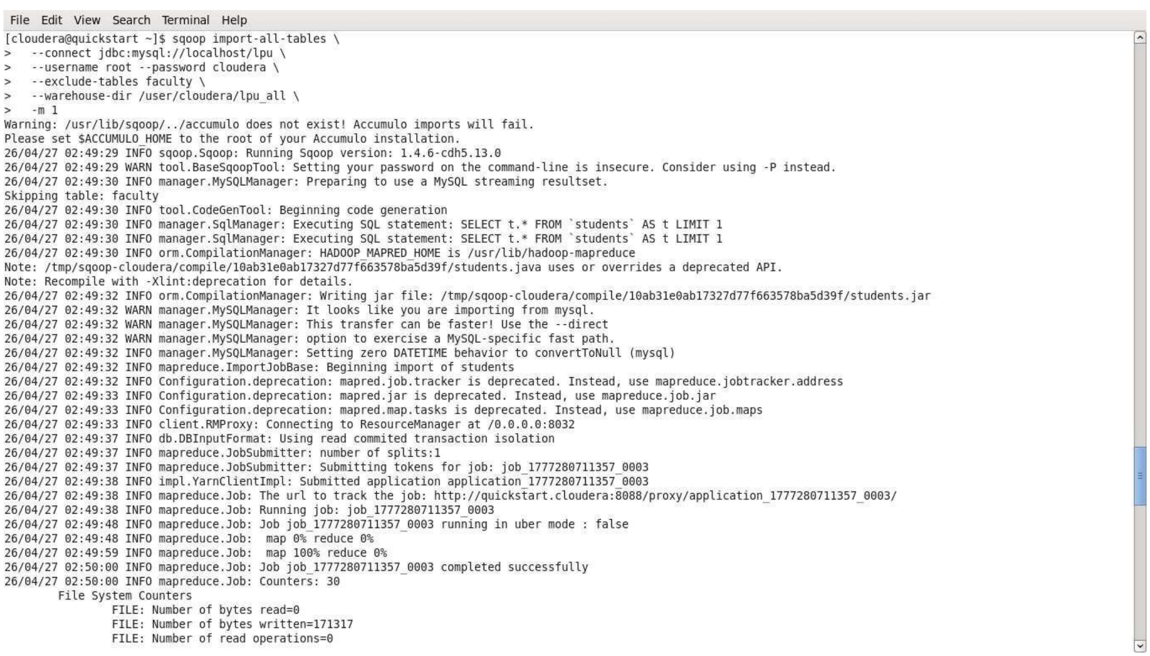
```
File  Edit  View  Search  Terminal  Help
26/04/27 02:46:31 INFO mapreduce.Job: Counters: 30
	File System Counters
		FILE: Number of bytes read=0
		FILE: Number of bytes written=171291
		FILE: Number of read operations=0
		FILE: Number of large read operations=0
		FILE: Number of write operations=0
		HDFS: Number of bytes read=87
		HDFS: Number of bytes written=125
		HDFS: Number of read operations=4
		HDFS: Number of large read operations=0
		HDFS: Number of write operations=2
	Job Counters
		Launched map tasks=1
		Other local map tasks=1
		Total time spent by all maps in occupied slots (ms)=10511
		Total time spent by all reduces in occupied slots (ms)=0
		Total time spent by all map tasks (ms)=10511
		Total vcore-milliseconds taken by all map tasks=10511
		Total megabyte-milliseconds taken by all map tasks=10763264
	Map-Reduce Framework
		Map input records=6
		Map output records=6
		Input split bytes=87
		Spilled Records=0
		Failed Shuffles=0
		Merged Map outputs=0
		GC time elapsed (ms)=117
		CPU time spent (ms)=1070
		Physical memory (bytes) snapshot=190484480
		Virtual memory (bytes) snapshot=1596346368
		Total committed heap usage (bytes)=173539328
	File Input Format Counters
		Bytes Read=0
	File Output Format Counters
		Bytes Written=125
26/04/27 02:46:31 INFO mapreduce.ImportJobBase: Transferred 125 bytes in 31.035 seconds (4.0277 bytes/sec)
26/04/27 02:46:31 INFO mapreduce.ImportJobBase: Retrieved 6 records.
[cloudera@quickstart ~]$ hdfs dfs -ls /user/cloudera/students
Found 2 items
-rw-r--r--   1 cloudera cloudera          0 2026-04-27 02:46 /user/cloudera/students/_SUCCESS
-rw-r--r--   1 cloudera cloudera        125 2026-04-27 02:46 /user/cloudera/students/part-m-00000
[cloudera@quickstart ~]$
```

Screenshot 5: Import complete — 6 records transferred. HDFS shows _SUCCESS and part-m-00000 at /user/cloudera/students.

Q2: Import All Tables Excluding Faculty

Command used: sqoop import-all-tables with --exclude-tables faculty and --warehouse-dir /user/cloudera/lpu_all

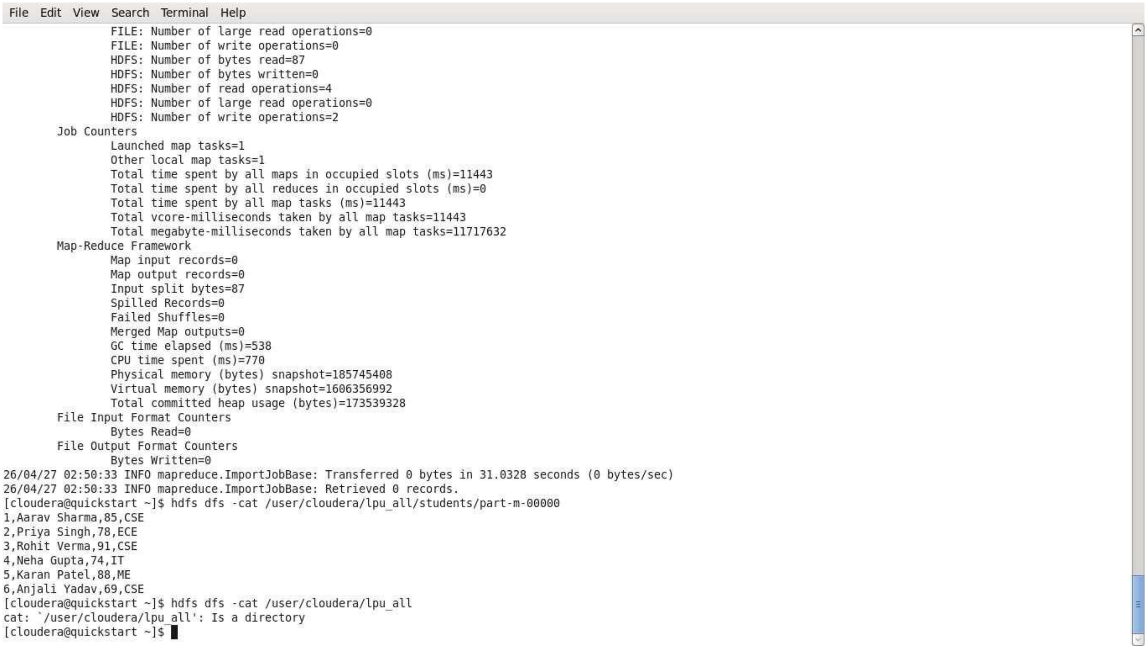
This imports every table in the lpu database except faculty. The output goes into /user/cloudera/lpu_all. You can see in the logs: "Skipping table: faculty".



```
File  Edit  View  Search  Terminal  Help
[cloudera@quickstart ~]$ sqoop import-all-tables \
>   --connect jdbc:mysql://localhost/lpu \
>   --username root --password cloudera \
>   --exclude-tables faculty \
>   --warehouse-dir /user/cloudera/lpu_all \
>   -m 1
Warning: /usr/lib/sqoop/../accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
26/04/27 02:49:29 INFO sqoop.Sqoop: Running Sqoop version: 1.4.6-cdh5.13.0
26/04/27 02:49:29 WARN tool.BaseSqoopTool: Setting your password on the command-line is insecure. Consider using -P instead.
26/04/27 02:49:30 INFO manager.MySQLManager: Preparing to use a MySQL streaming resultset.
Skipping table: faculty
26/04/27 02:49:30 INFO tool.CodeGenTool: Beginning code generation
26/04/27 02:49:30 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM `students` AS t LIMIT 1
26/04/27 02:49:30 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM `students` AS t LIMIT 1
26/04/27 02:49:30 INFO orm.CompilationManager: HADOOP_MAPRED_HOME is /usr/lib/hadoop-mapreduce
Note: /tmp/sqoop-cloudera/compile/10ab31e0ab17327d77f663578ba5d39f/students.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
26/04/27 02:49:32 INFO orm.CompilationManager: Writing jar file: /tmp/sqoop-cloudera/compile/10ab31e0ab17327d77f663578ba5d39f/students.jar
26/04/27 02:49:32 WARN manager.MySQLManager: It looks like you are importing from mysql.
26/04/27 02:49:32 WARN manager.MySQLManager: This transfer can be faster! Use the --direct
26/04/27 02:49:32 WARN manager.MySQLManager: option to exercise a MySQL-specific fast path.
26/04/27 02:49:32 INFO manager.MySQLManager: Setting zero DATETIME behavior to convertToNull (mysql)
26/04/27 02:49:32 INFO mapreduce.ImportJobBase: Beginning import of students
26/04/27 02:49:32 INFO Configuration.deprecation: mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
26/04/27 02:49:33 INFO Configuration.deprecation: mapred.jar is deprecated. Instead, use mapreduce.job.jar
26/04/27 02:49:33 INFO Configuration.deprecation: mapred.map.tasks is deprecated. Instead, use mapreduce.job.maps
26/04/27 02:49:33 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
26/04/27 02:49:37 INFO db.DBInputFormat: Using read commited transaction isolation
26/04/27 02:49:37 INFO mapreduce.JobSubmitter: number of splits:1
26/04/27 02:49:37 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1777280711357_0003
26/04/27 02:49:38 INFO impl.YarnClientImpl: Submitted application application_1777280711357_0003
26/04/27 02:49:38 INFO mapreduce.Job: The url to track the job: http://quickstart.cloudera:8088/proxy/application_1777280711357_0003/
26/04/27 02:49:38 INFO mapreduce.Job: Running job: job_1777280711357_0003
26/04/27 02:49:48 INFO mapreduce.Job: Job job_1777280711357_0003 running in uber mode : false
26/04/27 02:49:48 INFO mapreduce.Job:  map 0% reduce 0%
26/04/27 02:49:59 INFO mapreduce.Job:  map 100% reduce 0%
26/04/27 02:50:00 INFO mapreduce.Job: Job job_1777280711357_0003 completed successfully
26/04/27 02:50:00 INFO mapreduce.Job: Counters: 30
	File System Counters
		FILE: Number of bytes read=0
		FILE: Number of bytes written=171317
		FILE: Number of read operations=0
```

Screenshot 6: import-all-tables running. Sqoop skips faculty and imports students. MapReduce job submits.

Job finished. We ran hdfs dfs -cat to read the imported file and confirmed all 6 student records are there.



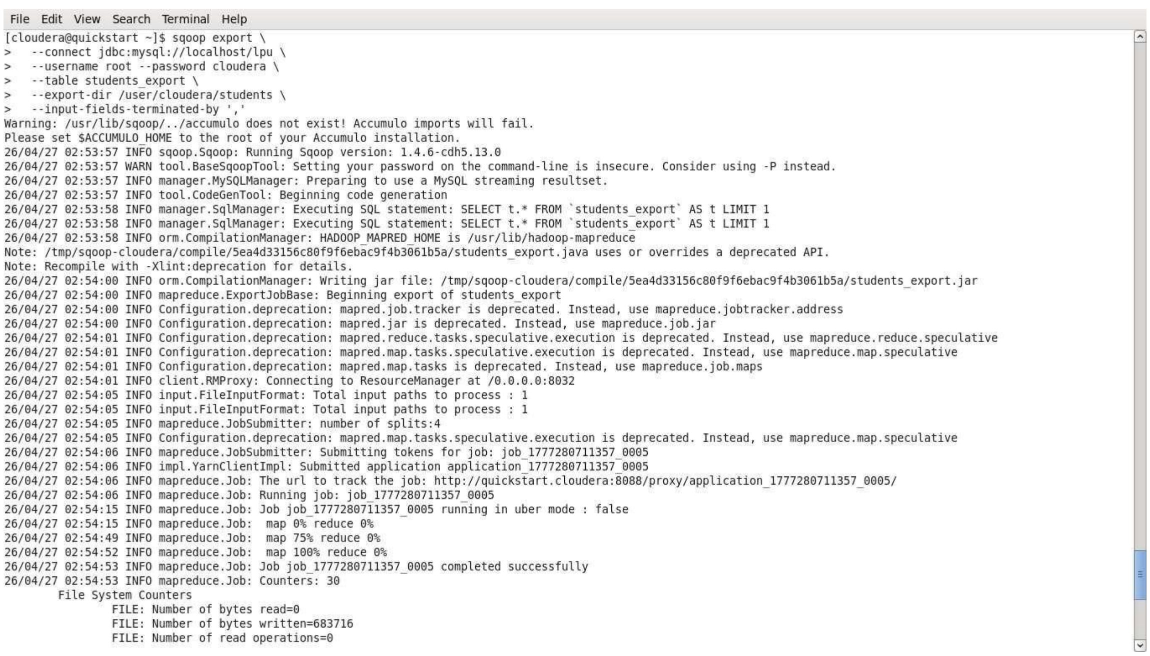
```
File  Edit  View  Search  Terminal  Help
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=87
                HDFS: Number of bytes written=0
                HDFS: Number of read operations=4
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
        Job Counters
                Launched map tasks=1
                Other local map tasks=1
                Total time spent by all maps in occupied slots (ms)=11443
                Total time spent by all reduces in occupied slots (ms)=0
                Total time spent by all map tasks (ms)=11443
                Total vcore-milliseconds taken by all map tasks=11443
                Total megabyte-milliseconds taken by all map tasks=11717632
        Map-Reduce Framework
                Map input records=0
                Map output records=0
                Input split bytes=87
                Spilled Records=0
                Failed Shuffles=0
                Merged Map outputs=0
                GC time elapsed (ms)=538
                CPU time spent (ms)=770
                Physical memory (bytes) snapshot=185745408
                Virtual memory (bytes) snapshot=1606356992
                Total committed heap usage (bytes)=173539328
        File Input Format Counters
                Bytes Read=0
        File Output Format Counters
                Bytes Written=0
26/04/27 02:50:33 INFO mapreduce.ImportJobBase: Transferred 0 bytes in 31.0328 seconds (0 bytes/sec)
26/04/27 02:50:33 INFO mapreduce.ImportJobBase: Retrieved 0 records.
[cloudera@quickstart ~]$ hdfs dfs -cat /user/cloudera/lpu_all/students/part-m-00000
1,Aarav Sharma,85,CSE
2,Priya Singh,78,ECE
3,Rohit Verma,91,CSE
4,Neha Gupta,74,IT
5,Karan Patel,88,ME
6,Anjali Yadav,69,CSE
[cloudera@quickstart ~]$ hdfs dfs -cat /user/cloudera/lpu_all
cat: `/user/cloudera/lpu_all': Is a directory
[cloudera@quickstart ~]$
```

Screenshot 7: Import done. cat on the file shows all 6 students in CSV format inside /user/cloudera/lpu_all/students/.

Q3: Export from HDFS Back to MySQL

Command used: sqoop export with --table students_export, --export-dir /user/cloudera/students, --input-fields-terminated-by ","

This reads the data from HDFS and pushes it into the MySQL students_export table. The delimiter tells Sqoop that fields in the HDFS file are separated by commas.



```
File  Edit  View  Search  Terminal  Help
[cloudera@quickstart ~]$ sqoop export \
>   --connect jdbc:mysql://localhost/lpu \
>   --username root --password cloudera \
>   --table students_export \
>   --export-dir /user/cloudera/students \
>   --input-fields-terminated-by ','
Warning: /usr/lib/sqoop/../accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
26/04/27 02:53:57 INFO sqoop.Sqoop: Running Sqoop version: 1.4.6-cdh5.13.0
26/04/27 02:53:57 WARN tool.BaseSqoopTool: Setting your password on the command-line is insecure. Consider using -P instead.
26/04/27 02:53:57 INFO manager.MySQLManager: Preparing to use a MySQL streaming resultset.
26/04/27 02:53:57 INFO tool.CodeGenTool: Beginning code generation
26/04/27 02:53:58 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM `students_export` AS t LIMIT 1
26/04/27 02:53:58 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM `students_export` AS t LIMIT 1
26/04/27 02:53:58 INFO orm.CompilationManager: HADOOP_MAPRED_HOME is /usr/lib/hadoop-mapreduce
Note: /tmp/sqoop-cloudera/compile/5ea4d33156c80f9f6ebac9f4b3061b5a/students_export.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
26/04/27 02:54:00 INFO orm.CompilationManager: Writing jar file: /tmp/sqoop-cloudera/compile/5ea4d33156c80f9f6ebac9f4b3061b5a/students_export.jar
26/04/27 02:54:00 INFO mapreduce.ExportJobBase: Beginning export of students_export
26/04/27 02:54:00 INFO Configuration.deprecation: mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
26/04/27 02:54:00 INFO Configuration.deprecation: mapred.jar is deprecated. Instead, use mapreduce.job.jar
26/04/27 02:54:01 INFO Configuration.deprecation: mapred.reduce.tasks.speculative.execution is deprecated. Instead, use mapreduce.reduce.speculative
26/04/27 02:54:01 INFO Configuration.deprecation: mapred.map.tasks.speculative.execution is deprecated. Instead, use mapreduce.map.speculative
26/04/27 02:54:01 INFO Configuration.deprecation: mapred.map.tasks is deprecated. Instead, use mapreduce.job.maps
26/04/27 02:54:01 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
26/04/27 02:54:05 INFO input.FileInputFormat: Total input paths to process : 1
26/04/27 02:54:05 INFO input.FileInputFormat: Total input paths to process : 1
26/04/27 02:54:05 INFO mapreduce.JobSubmitter: number of splits:4
26/04/27 02:54:05 INFO Configuration.deprecation: mapred.map.tasks.speculative.execution is deprecated. Instead, use mapreduce.map.speculative
26/04/27 02:54:06 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1777280711357_0005
26/04/27 02:54:06 INFO impl.YarnClientImpl: Submitted application application_1777280711357_0005
26/04/27 02:54:06 INFO mapreduce.Job: The url to track the job: http://quickstart.cloudera:8088/proxy/application_1777280711357_0005/
26/04/27 02:54:06 INFO mapreduce.Job: Running job: job_1777280711357_0005
26/04/27 02:54:15 INFO mapreduce.Job: Job job_1777280711357_0005 running in uber mode : false
26/04/27 02:54:15 INFO mapreduce.Job:  map 0% reduce 0%
26/04/27 02:54:49 INFO mapreduce.Job:  map 75% reduce 0%
26/04/27 02:54:52 INFO mapreduce.Job:  map 100% reduce 0%
26/04/27 02:54:53 INFO mapreduce.Job: Job job_1777280711357_0005 completed successfully
26/04/27 02:54:53 INFO mapreduce.Job: Counters: 30
        File System Counters
                FILE: Number of bytes read=0
                FILE: Number of bytes written=683716
                FILE: Number of read operations=0
```

Screenshot 8: Sqoop export job running. 4 splits used, map goes from 0% to 75% to 100%.

Export finished — 6 records exported. We logged into MySQL and did SELECT * FROM students_export to verify the data came through correctly.

```
File  Edit  View  Search  Terminal  Help
                GC time elapsed (ms)=1682
                CPU time spent (ms)=3590
                Physical memory (bytes) snapshot=684089344
                Virtual memory (bytes) snapshot=6336348160
                Total committed heap usage (bytes)=630194176
        File Input Format Counters
                Bytes Read=0
        File Output Format Counters
                Bytes Written=0
26/04/27 02:54:53 INFO mapreduce.ExportJobBase: Transferred 1,021 bytes in 52.3382 seconds (19.5078 bytes/sec)
26/04/27 02:54:53 INFO mapreduce.ExportJobBase: Exported 6 records.
[cloudera@quickstart ~]$ mysql -u root -pcloudera
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 34
Server version: 5.1.73 Source distribution

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Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE lpu;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SELECT * FROM students_export;
+----+--------------+-------+--------+
| id | name         | marks | branch |
+----+--------------+-------+--------+
|  1 | Aarav Sharma |    85 | CSE    |
|  2 | Priya Singh  |    78 | ECE    |
|  3 | Rohit Verma  |    91 | CSE    |
|  4 | Neha Gupta   |    74 | IT     |
|  5 | Karan Patel  |    88 | ME     |
|  6 | Anjali Yadav |    69 | CSE    |
+----+--------------+-------+--------+
6 rows in set (0.00 sec)

mysql> 
```

Screenshot 9: Export done. MySQL shows all 6 students in students_export — exact same data as the original students table.

TASK 2 — PIG: Analytics Commands

Setup: Create and Upload students.txt to HDFS

We created students.txt on the Desktop using cat command and typed in 6 records (one per line, comma separated). Then uploaded it to HDFS at /user/cloudera/students.txt using hdfs dfs -put. After that we launched the pig shell.

```
File  Edit  View  Search  Terminal  Help
[cloudera@quickstart ~]$ cat > /home/cloudera/students.txt
^C
[cloudera@quickstart ~]$ clear
[cloudera@quickstart ~]$ cat > /home/cloudera/Desktop/students.txt
1,Aarav Sharma,85,CSE
2,Priya Singh,78,ECE
3,Rohit Verma,91,CSE
4,Neha Gupta,74,IT
5,Karan Patel,88,ME
6,Anjali Yadav,69,CSE
^C
[cloudera@quickstart ~]$ hdfs dfs -put /home/cloudera/Desktop/students.txt /user
/cloudera/
[cloudera@quickstart ~]$ hdfs dfs -ls /user/cloudera/
Found 12 items
-rw-r--r--   1 cloudera cloudera         26 2026-04-23 03:38 /user/cloudera/empty
drwxr-xr-x   - cloudera cloudera          0 2026-04-27 02:50 /user/cloudera/lpu_all
drwxr-xr-x   - cloudera cloudera          0 2026-04-23 02:52 /user/cloudera/mark
drwxr-xr-x   - cloudera cloudera          0 2026-04-23 02:09 /user/cloudera/mba
drwxr-xr-x   - cloudera cloudera          0 2026-04-23 02:24 /user/cloudera/mba2
drwxr-xr-x   - cloudera cloudera          0 2026-04-23 02:46 /user/cloudera/mba3
drwxr-xr-x   - cloudera cloudera          0 2026-04-23 02:52 /user/cloudera/msb
drwxr-xr-x   - cloudera cloudera          0 2026-04-23 03:04 /user/cloudera/msb3
drwxr-xr-x   - cloudera cloudera          0 2026-04-21 03:43 /user/cloudera/ques1.2
drwxr-xr-x   - cloudera cloudera          0 2026-04-21 02:54 /user/cloudera/serverdir
drwxr-xr-x   - cloudera cloudera          0 2026-04-27 02:46 /user/cloudera/students
-rw-r--r--   1 cloudera cloudera        125 2026-04-27 03:10 /user/cloudera/students.txt
[cloudera@quickstart ~]$ pig
log4j:WARN No appenders could be found for logger (org.apache.hadoop.util.Shell).
log4j:WARN Please initialize the log4j system properly.
log4j:WARN See http://logging.apache.org/log4j/1.2/faq.html#noconfig for more info.
2026-04-27 03:11:32,521 [main] INFO  org.apache.pig.Main - Apache Pig version 0.12.0-cdh5.13.0 (rexported) compiled Oct 04 2017, 11:09:03
2026-04-27 03:11:32,521 [main] INFO  org.apache.pig.Main - Logging error messages to: /home/cloudera/pig_1777284692501.log
2026-04-27 03:11:32,545 [main] INFO  org.apache.pig.impl.util.Utils - Default bootup file /home/cloudera/.pigbootup not found
2026-04-27 03:11:33,234 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-04-27 03:11:33,234 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated. Instead, use fs.defaultFS
2026-04-27 03:11:33,234 [main] INFO  org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://quickstart.cloudera:80
20
2026-04-27 03:11:34,683 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-04-27 03:11:34,683 [main] INFO  org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to map-reduce job tracker at: localhost:8021
2026-04-27 03:11:34,684 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated. Instead, use fs.defaultFS
2026-04-27 03:11:34,713 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated. Instead, use fs.defaultFS
2026-04-27 03:11:34,713 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
```

Screenshot 1: students.txt created, uploaded to HDFS, confirmed with hdfs dfs -ls. Pig shell launched.

Command 1: LOAD — Load Data into a Bag

Command: data = LOAD '/user/cloudera/students.txt' USING PigStorage(',') AS (id:int, name:chararray, marks:int, branch:chararray);

This reads the CSV file from HDFS and maps each column to a name and type. The result is stored in a relation called data. DUMP data shows all 6 records.

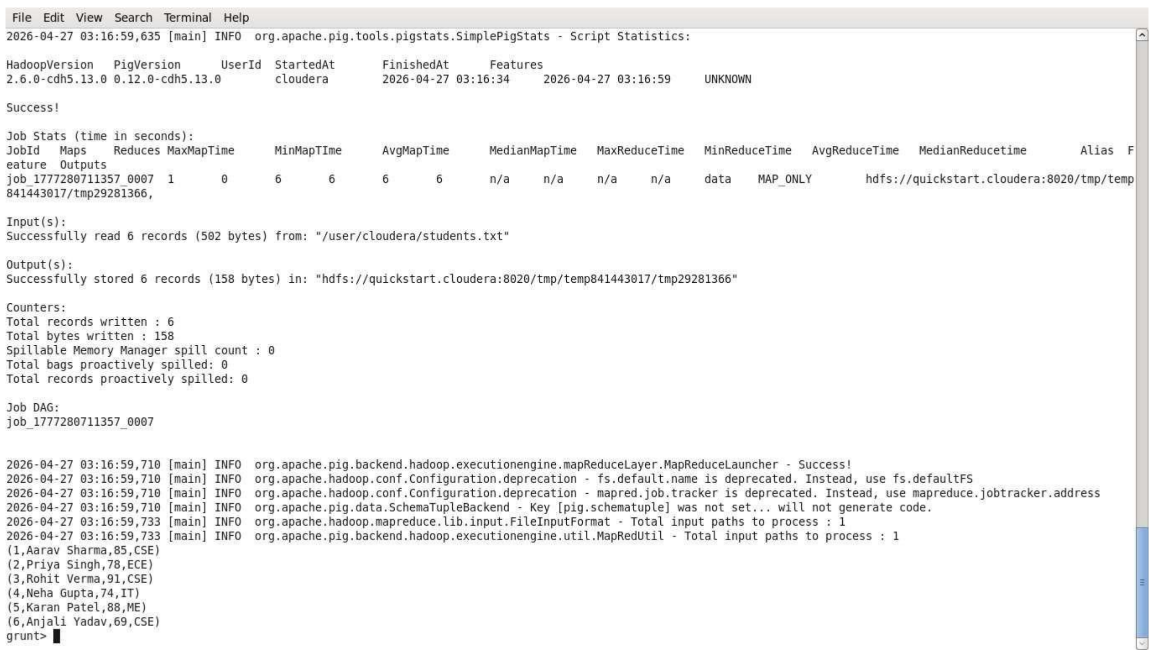
```
File  Edit  View  Search  Terminal  Help
grunt> data = LOAD '/user/cloudera/students.txt' USING PigStorage(',') AS (id:int, name:chararray, marks:int, branch:chararray);
grunt> DUMP data;
2026-04-27 03:13:54,808 [main] INFO  org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: UNKNOWN
2026-04-27 03:13:54,870 [main] INFO  org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, DuplicateForEachColu
mnRewrite, GroupByConstParallelSetter, ImplicitSplitInserter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NewPartitionFilterOptimizer, PushDownForE
achFlatten, PushUpFilter, SplitFilter, StreamTypeCastInserter], RULES_DISABLED=[FilterLogicExpressionSimplifier, PartitionFilterOptimizer]}
2026-04-27 03:13:55,084 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-04-27 03:13:55,183 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2026-04-27 03:13:55,183 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2026-04-27 03:13:55,858 [main] INFO  org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-04-27 03:13:56,469 [main] INFO  org.apache.pig.tools.pigstats.ScriptState - Pig script settings are added to the job
2026-04-27 03:13:56,557 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce
.reduce.markreset.buffer.percent
2026-04-27 03:13:56,557 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not
 set, set to default 0.3
2026-04-27 03:13:56,557 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputfo
rmat.compress
2026-04-27 03:13:58,104 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - creating jar file Job748545268951916960.jar
2026-04-27 03:14:00,682 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - jar file Job748545268951916960.jar created
2026-04-27 03:14:00,682 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.jar is deprecated. Instead, use mapreduce.job.jar
2026-04-27 03:14:00,705 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting up single store job
2026-04-27 03:14:00,711 [main] INFO  org.apache.pig.data.SchemaTupleFrontend - Key [pig.schematuple] is false, will not generate code.
2026-04-27 03:14:00,711 [main] INFO  org.apache.pig.data.SchemaTupleFrontend - Starting process to move generated code to distributed cache
2026-04-27 03:14:00,711 [main] INFO  org.apache.pig.data.SchemaTupleFrontend - Setting key [pig.schematuple.classes] with classes to deserialize []
2026-04-27 03:14:00,737 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
2026-04-27 03:14:00,738 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker.http.address is deprecated. Instead, use mapreduce.jobtracker
.http.address
2026-04-27 03:14:00,738 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-04-27 03:14:00,754 [JobControl] INFO  org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-04-27 03:14:00,779 [JobControl] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated. Instead, use fs.defaultFS
2026-04-27 03:14:01,237 [JobControl] INFO  org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input paths to process : 1
2026-04-27 03:14:01,237 [JobControl] INFO  org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
2026-04-27 03:14:01,264 [JobControl] INFO  org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths (combined) to process : 1
2026-04-27 03:14:01,743 [JobControl] INFO  org.apache.hadoop.mapreduce.JobSubmitter - number of splits:1
2026-04-27 03:14:01,936 [JobControl] INFO  org.apache.hadoop.mapreduce.JobSubmitter - Submitting tokens for job: job_1777280711357_0006
2026-04-27 03:14:02,401 [JobControl] INFO  org.apache.hadoop.yarn.client.api.impl.YarnClientImpl - Submitted application application_1777280711357_0006
2026-04-27 03:14:02,446 [JobControl] INFO  org.apache.hadoop.mapreduce.Job - The url to track the job: http://quickstart.cloudera:8088/proxy/application_1777280711357_0
006/
2026-04-27 03:14:02,447 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - HadoopJobId: job_1777280711357_0006
2026-04-27 03:14:02,447 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Processing aliases data
2026-04-27 03:14:02,447 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - detailed locations: M: data[1,7],data[-1,-1] C:  R
:
2026-04-27 03:14:02,447 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - More information at: http://localhost:50030/jobdet
```

Screenshot 2: LOAD command runs. MapReduce job fires up to process the file.

```
File  Edit  View  Search  Terminal  Help
2026-04-27 03:14:32,807 [main] INFO  org.apache.pig.tools.pigstats.SimplePigStats - Script Statistics:

HadoopVersion   PigVersion      UserId  StartedAt       FinishedAt      Features
2.6.0-cdh5.13.0 0.12.0-cdh5.13.0        cloudera        2026-04-27 03:13:56     2026-04-27 03:14:32     UNKNOWN

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTIme      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReducetime        Alias   F
eature  Outputs
job_1777280711357_0006  1       0       11      11      11      11      n/a     n/a     n/a     n/a     data    MAP_ONLY        hdfs://quickstart.cloudera:8020/tmp/temp
841443017/tmp664913659,

Input(s):
Successfully read 6 records (502 bytes) from: "/user/cloudera/students.txt"

Output(s):
Successfully stored 6 records (158 bytes) in: "hdfs://quickstart.cloudera:8020/tmp/temp841443017/tmp664913659"

Counters:
Total records written : 6
Total bytes written : 158
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1777280711357_0006


2026-04-27 03:14:32,881 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-04-27 03:14:32,889 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated. Instead, use fs.defaultFS
2026-04-27 03:14:32,889 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-04-27 03:14:32,890 [main] INFO  org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-04-27 03:14:32,921 [main] INFO  org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input paths to process : 1
2026-04-27 03:14:32,923 [main] INFO  org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
(1,Aarav Sharma,85,CSE)
(2,Priya Singh,78,ECE)
(3,Rohit Verma,91,CSE)
(4,Neha Gupta,74,IT)
(5,Karan Patel,88,ME)
(6,Anjali Yadav,69,CSE)
grunt>
```

Screenshot 3: DUMP result — all 6 student records loaded correctly from the file.

Screenshot 4: LOAD run again in a fresh session before running DUMP.



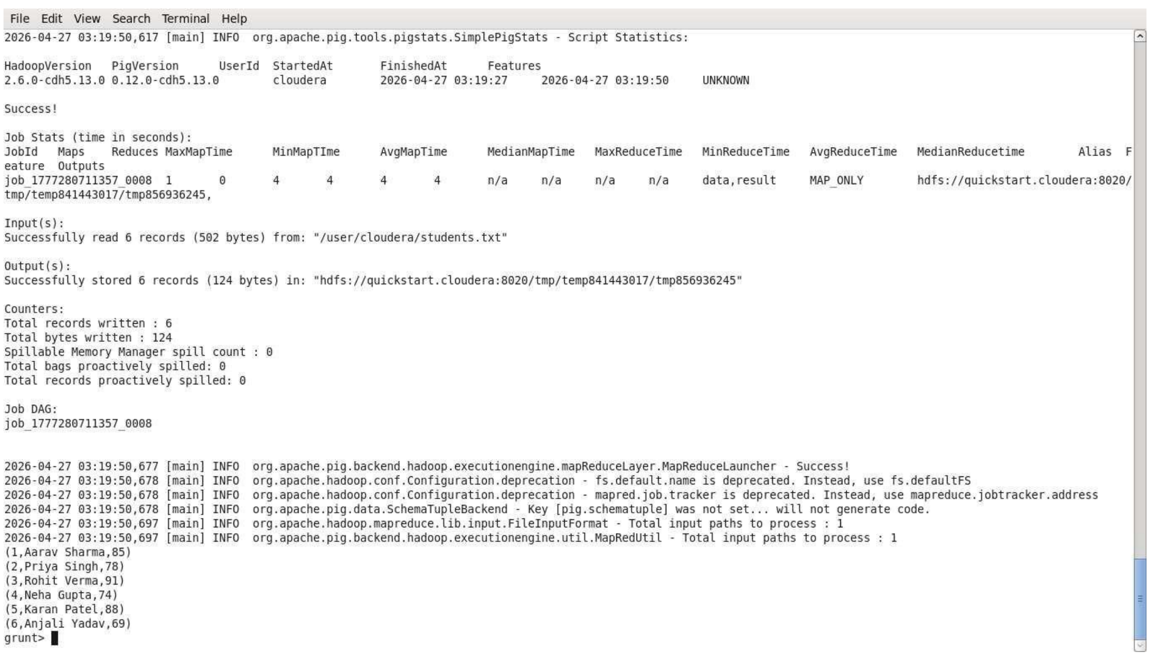
Screenshot 5: DUMP data confirmed — 6 rows with id, name, marks, branch.

Command 2: FOREACH — Select Specific Columns

Command: result = FOREACH data GENERATE id, name, marks;

FOREACH goes through every row in data and picks only id, name, and marks — dropping the branch column. Like a SELECT in SQL.

Screenshot 6: FOREACH command runs with MapReduce job.



Screenshot 7: DUMP result shows 6 rows with only id, name, marks — branch column removed.

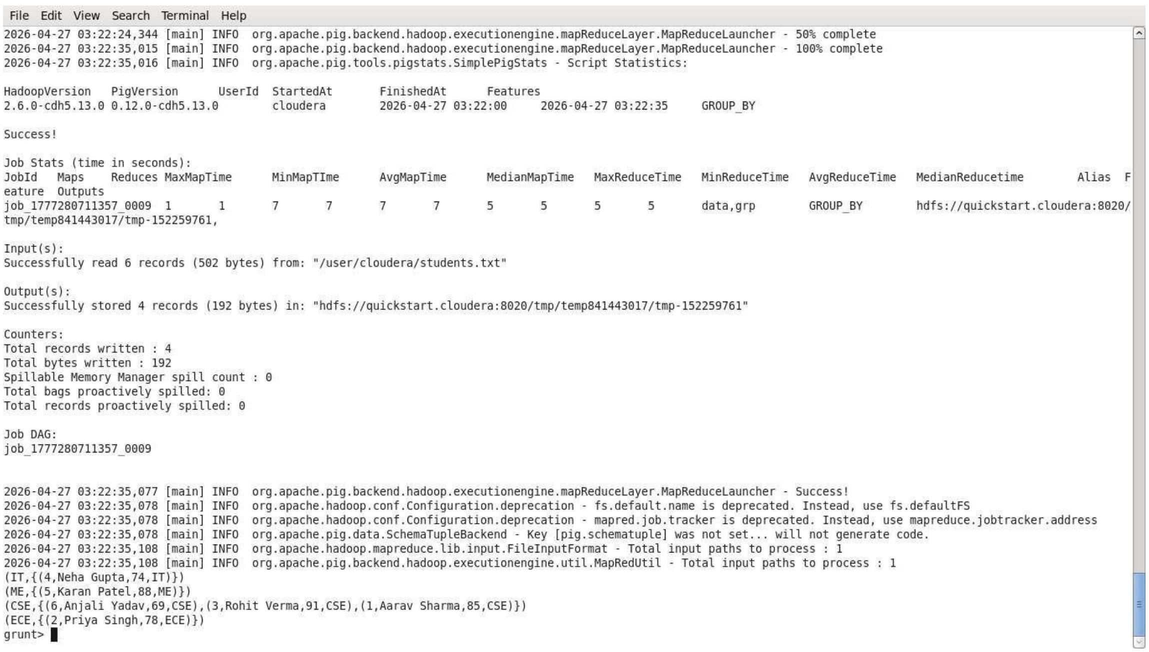
Command 3: GROUP BY — Group Students by Branch

Command: grp = GROUP data BY branch;

This groups all students by their branch. Students in the same branch get collected into a bag together. CSE gets 3 students, ECE gets 1, IT gets 1, ME gets 1.

```
File  Edit  View  Search  Terminal  Help
grunt> grp = GROUP data BY branch;
grunt> DUMP grp;
2026-04-27 03:22:00,134 [main] INFO  org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY
2026-04-27 03:22:00,138 [main] INFO  org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, DuplicateForEachColu
mnRewrite, GroupByConstParallelSetter, ImplicitSplitInserter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NewPartitionFilterOptimizer, PushDownForE
achFlatten, PushUpFilter, SplitFilter, StreamTypeCastInserter], RULES_DISABLED=[FilterLogicExpressionSimplifier, PartitionFilterOptimizer]}
2026-04-27 03:22:00,174 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-04-27 03:22:00,201 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2026-04-27 03:22:00,201 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2026-04-27 03:22:00,281 [main] INFO  org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-04-27 03:22:00,293 [main] INFO  org.apache.pig.tools.pigstats.ScriptState - Pig script settings are added to the job
2026-04-27 03:22:00,305 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not
 set, set to default 0.3
2026-04-27 03:22:00,305 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required r
educers.
2026-04-27 03:22:00,314 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Using reducer estimator: org.apache.pig.backend.h
adoop.executionengine.mapReduceLayer.InputSizeReducerEstimator
2026-04-27 03:22:00,320 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator - BytesPerReducer=1000000000 maxReducers=999
 totalInputFileSize=125
2026-04-27 03:22:00,320 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting Parallelism to 1
2026-04-27 03:22:01,408 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - creating jar file Job73707435923518934.jar
2026-04-27 03:22:04,118 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - jar file Job73707435923518934.jar created
2026-04-27 03:22:04,128 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting up single store job
2026-04-27 03:22:04,129 [main] INFO  org.apache.pig.data.SchemaTupleFrontend - Key [pig.schematuple] is false, will not generate code.
2026-04-27 03:22:04,129 [main] INFO  org.apache.pig.data.SchemaTupleFrontend - Starting process to move generated code to distributed cache
2026-04-27 03:22:04,129 [main] INFO  org.apache.pig.data.SchemaTupleFrontend - Setting key [pig.schematuple.classes] with classes to deserialize []
2026-04-27 03:22:04,224 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
2026-04-27 03:22:04,225 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-04-27 03:22:04,248 [JobControl] INFO  org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-04-27 03:22:04,258 [JobControl] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated. Instead, use fs.defaultFS
2026-04-27 03:22:04,471 [JobControl] INFO  org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input paths to process : 1
2026-04-27 03:22:04,471 [JobControl] INFO  org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
2026-04-27 03:22:04,476 [JobControl] INFO  org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths (combined) to process : 1
2026-04-27 03:22:04,528 [JobControl] INFO  org.apache.hadoop.mapreduce.JobSubmitter - number of splits:1
2026-04-27 03:22:04,614 [JobControl] INFO  org.apache.hadoop.mapreduce.JobSubmitter - Submitting tokens for job: job_1777280711357_0009
2026-04-27 03:22:04,722 [JobControl] INFO  org.apache.hadoop.yarn.client.api.impl.YarnClientImpl - Submitted application application_1777280711357_0009
2026-04-27 03:22:04,735 [JobControl] INFO  org.apache.hadoop.mapreduce.Job - The url to track the job: http://quickstart.cloudera:8088/proxy/application_1777280711357_0
009/
2026-04-27 03:22:04,735 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - HadoopJobId: job_1777280711357_0009
2026-04-27 03:22:04,735 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Processing aliases data,grp
2026-04-27 03:22:04,735 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - detailed locations: M: data[2,7],data[-1,-1],grp[4
,6] C:  R:
2026-04-27 03:22:04,735 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - More information at: http://localhost:50030/jobdet
```

Screenshot 8: GROUP BY command runs. Pig fires a MapReduce job with a reduce phase this time.



```
File  Edit  View  Search  Terminal  Help
2026-04-27 03:22:24,344 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 50% complete
2026-04-27 03:22:35,015 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 100% complete
2026-04-27 03:22:35,016 [main] INFO  org.apache.pig.tools.pigstats.SimplePigStats - Script Statistics:

HadoopVersion   PigVersion      UserId  StartedAt       FinishedAt      Features
2.6.0-cdh5.13.0 0.12.0-cdh5.13.0        cloudera        2026-04-27 03:22:00     2026-04-27 03:22:35     GROUP_BY

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTIme      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReducetime        Alias   F
eature  Outputs
job_1777280711357_0009  1       1       7       7       7       7       5       5       5       5       data,grp        GROUP_BY        hdfs://quickstart.cloudera:8020/
tmp/temp841443017/tmp-152259761,

Input(s):
Successfully read 6 records (502 bytes) from: "/user/cloudera/students.txt"

Output(s):
Successfully stored 4 records (192 bytes) in: "hdfs://quickstart.cloudera:8020/tmp/temp841443017/tmp-152259761"

Counters:
Total records written : 4
Total bytes written : 192
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1777280711357_0009


2026-04-27 03:22:35,077 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-04-27 03:22:35,078 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated. Instead, use fs.defaultFS
2026-04-27 03:22:35,078 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-04-27 03:22:35,078 [main] INFO  org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-04-27 03:22:35,108 [main] INFO  org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input paths to process : 1
2026-04-27 03:22:35,108 [main] INFO  org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
(IT,{(4,Neha Gupta,74,IT)})
(ME,{(5,Karan Patel,88,ME)})
(CSE,{(6,Anjali Yadav,69,CSE),(3,Rohit Verma,91,CSE),(1,Aarav Sharma,85,CSE)})
(ECE,{(2,Priya Singh,78,ECE)})
grunt>
```

Screenshot 9: DUMP grp — 4 groups: IT, ME, CSE (3 students), ECE. Each group shows its bag of records.

Command 4: ORDER BY — Sort by Marks Descending

Command: ordered = ORDER data BY marks DESC;

This sorts all 6 students from highest to lowest marks. Rohit (91) comes first, Anjali (69) comes last.

Screenshot 10: ORDER BY command runs. Multiple MapReduce jobs fire to handle the sort.

Screenshot 11: DUMP ordered — students sorted by marks descending: 91, 88, 85, 78, 74, 69.

Command 5: FILTER — Keep Only Marks > 75

Command: pass = FILTER data BY marks > 75;

This filters out students who scored 75 or below. Only 4 students pass: Aarav (85), Priya (78), Rohit (91), Karan (88). Neha (74) and Anjali (69) are excluded.

```
File  Edit  View  Search  Terminal  Help
grunt> pass = FILTER data BY marks > 75;
grunt> DUMP pass;
2026-04-27 03:26:40,273 [main] INFO  org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: FILTER
2026-04-27 03:26:40,277 [main] INFO  org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, DuplicateForEachColu
mnRewrite, GroupByConstParallelSetter, ImplicitSplitInserter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NewPartitionFilterOptimizer, PushDownForE
achFlatten, PushUpFilter, SplitFilter, StreamTypeCastInserter], RULES_DISABLED=[FilterLogicExpressionSimplifier, PartitionFilterOptimizer]}
2026-04-27 03:26:40,293 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-04-27 03:26:40,296 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2026-04-27 03:26:40,296 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2026-04-27 03:26:40,331 [main] INFO  org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-04-27 03:26:40,338 [main] INFO  org.apache.pig.tools.pigstats.ScriptState - Pig script settings are added to the job
2026-04-27 03:26:40,353 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not
 set, set to default 0.3
2026-04-27 03:26:42,057 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - creating jar file Job34035212790869889.jar
2026-04-27 03:26:44,680 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - jar file Job34035212790869889.jar created
2026-04-27 03:26:44,697 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting up single store job
2026-04-27 03:26:44,699 [main] INFO  org.apache.pig.data.SchemaTupleFrontend - Key [pig.schematuple] is false, will not generate code.
2026-04-27 03:26:44,699 [main] INFO  org.apache.pig.data.SchemaTupleFrontend - Starting process to move generated code to distributed cache
2026-04-27 03:26:44,699 [main] INFO  org.apache.pig.data.SchemaTupleFrontend - Setting key [pig.schematuple.classes] with classes to deserialize []
2026-04-27 03:26:44,720 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
2026-04-27 03:26:44,722 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-04-27 03:26:44,779 [JobControl] INFO  org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-04-27 03:26:44,789 [JobControl] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated. Instead, use fs.defaultFS
2026-04-27 03:26:44,920 [JobControl] INFO  org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input paths to process : 1
2026-04-27 03:26:44,920 [JobControl] INFO  org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
2026-04-27 03:26:44,923 [JobControl] INFO  org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths (combined) to process : 1
2026-04-27 03:26:45,361 [JobControl] INFO  org.apache.hadoop.mapreduce.JobSubmitter - number of splits:1
2026-04-27 03:26:45,419 [JobControl] INFO  org.apache.hadoop.mapreduce.JobSubmitter - Submitting tokens for job: job_1777280711357_0013
2026-04-27 03:26:45,485 [JobControl] INFO  org.apache.hadoop.yarn.client.api.impl.YarnClientImpl - Submitted application application_1777280711357_0013
2026-04-27 03:26:45,492 [JobControl] INFO  org.apache.hadoop.mapreduce.Job - The url to track the job: http://quickstart.cloudera:8088/proxy/application_1777280711357_0
013/
2026-04-27 03:26:45,493 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - HadoopJobId: job_1777280711357_0013
2026-04-27 03:26:45,493 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Processing aliases data,pass
2026-04-27 03:26:45,493 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - detailed locations: M: data[2,7],data[-1,-1],pass[
6,7] C:  R:
2026-04-27 03:26:45,493 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - More information at: http://localhost:50030/jobdet
ails.jsp?jobid=job_1777280711357_0013
2026-04-27 03:26:45,558 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 0% complete
2026-04-27 03:27:00,670 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 50% complete
2026-04-27 03:27:06,155 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 100% complete
2026-04-27 03:27:06,157 [main] INFO  org.apache.pig.tools.pigstats.SimplePigStats - Script Statistics:

HadoopVersion   PigVersion      UserId  StartedAt       FinishedAt      Features
```

Screenshot 12: FILTER command runs.

```
File  Edit  View  Search  Terminal  Help
2026-04-27 03:27:00,670 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 50% complete
2026-04-27 03:27:06,155 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 100% complete
2026-04-27 03:27:06,157 [main] INFO  org.apache.pig.tools.pigstats.SimplePigStats - Script Statistics:

HadoopVersion   PigVersion      UserId  StartedAt       FinishedAt      Features
2.6.0-cdh5.13.0 0.12.0-cdh5.13.0        cloudera        2026-04-27 03:26:40     2026-04-27 03:27:06     FILTER

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTIme      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReducetime        Alias   F
eature  Outputs
job_1777280711357_0013  1       0       6       6       6       6       n/a     n/a     n/a     n/a     data,pass       MAP_ONLY        hdfs://quickstart.cloudera:8020/
tmp/temp841443017/tmp366125317,

Input(s):
Successfully read 6 records (502 bytes) from: "/user/cloudera/students.txt"

Output(s):
Successfully stored 4 records (109 bytes) in: "hdfs://quickstart.cloudera:8020/tmp/temp841443017/tmp366125317"

Counters:
Total records written : 4
Total bytes written : 109
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1777280711357_0013


2026-04-27 03:27:06,218 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-04-27 03:27:06,219 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated. Instead, use fs.defaultFS
2026-04-27 03:27:06,219 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-04-27 03:27:06,219 [main] INFO  org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-04-27 03:27:06,262 [main] INFO  org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input paths to process : 1
2026-04-27 03:27:06,262 [main] INFO  org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
(1,Aarav Sharma,85,CSE)
(2,Priya Singh,78,ECE)
(3,Rohit Verma,91,CSE)
(5,Karan Patel,88,ME)
grunt>
```

Screenshot 13: DUMP pass — 4 records where marks > 75. Neha and Anjali are correctly filtered out.

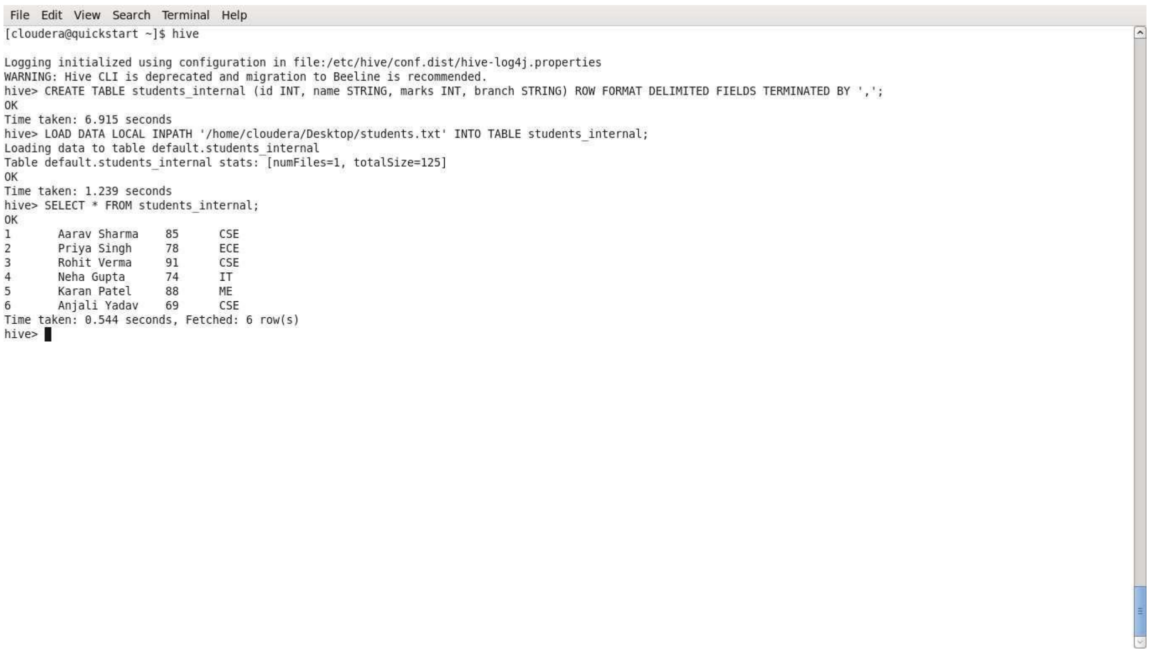
TASK 3 — HIVE: Table Operations

1. Internal Table — students_internal

Command: CREATE TABLE students_internal (id INT, name STRING, marks INT, branch STRING) ROW FORMAT DELIMITED FIELDS TERMINATED BY ',';

Then: LOAD DATA LOCAL INPATH '/home/cloudera/Desktop/students.txt' INTO TABLE students_internal;

An internal (managed) table means Hive owns the data. If you drop the table, the data is deleted too. We loaded the local file directly into it.

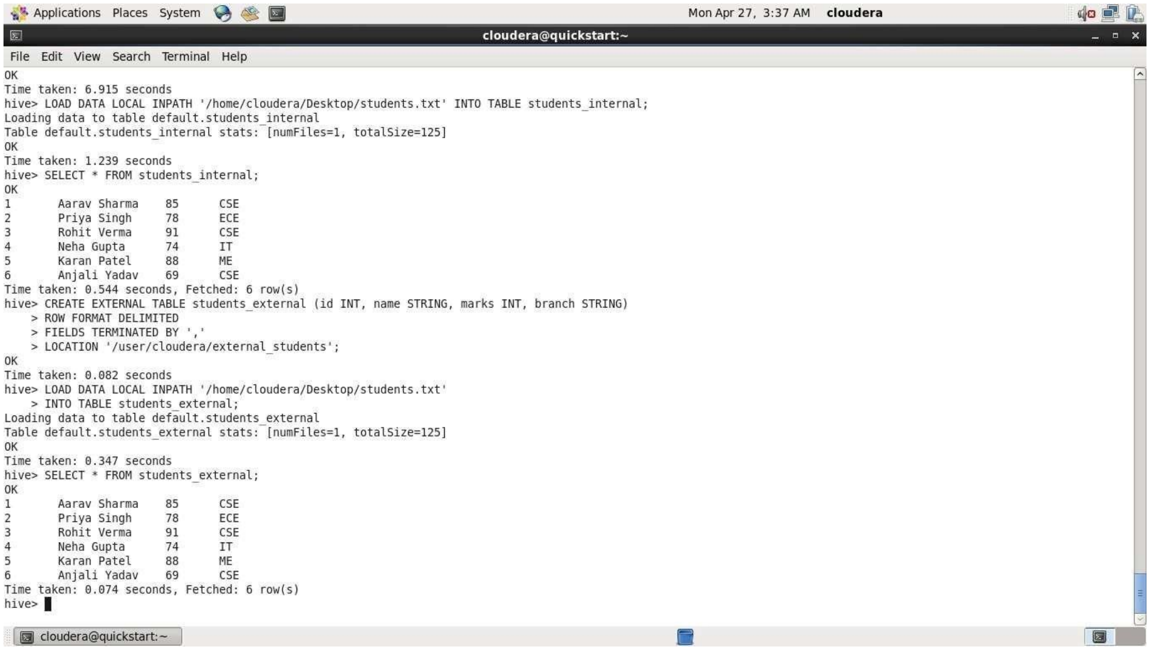


*Screenshot 1: Internal table created. File loaded. SELECT * shows all 6 students.*

2. External Table — students_external

Command: CREATE EXTERNAL TABLE students_external (id INT, name STRING, marks INT, branch STRING) ROW FORMAT DELIMITED FIELDS TERMINATED BY ',' LOCATION '/user/cloudera/external_students';

An external table means Hive just points to data in HDFS. If you drop the table, the actual data in HDFS stays safe. Good when multiple tools need to access the same data.

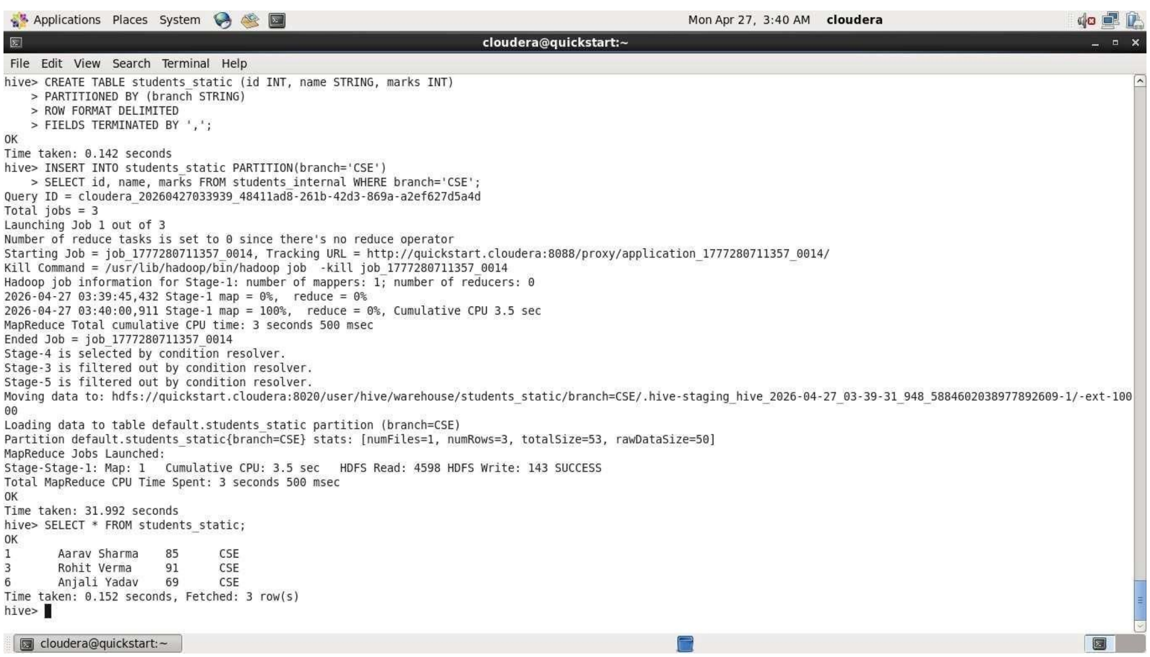


*Screenshot 2: External table created with HDFS location. Data loaded. SELECT * confirms all 6 rows.*

3. Static Partition — students_static

Commands: CREATE TABLE students_static (id INT, name STRING, marks INT) PARTITIONED BY (branch STRING). Then INSERT INTO students_static PARTITION(branch='CSE') SELECT id, name, marks FROM students_internal WHERE branch='CSE';

Static partitioning means you manually tell Hive which partition to write into. We pushed only the CSE students (Aarav, Rohit, Anjali) into the CSE partition. The SELECT result shows those 3 rows.

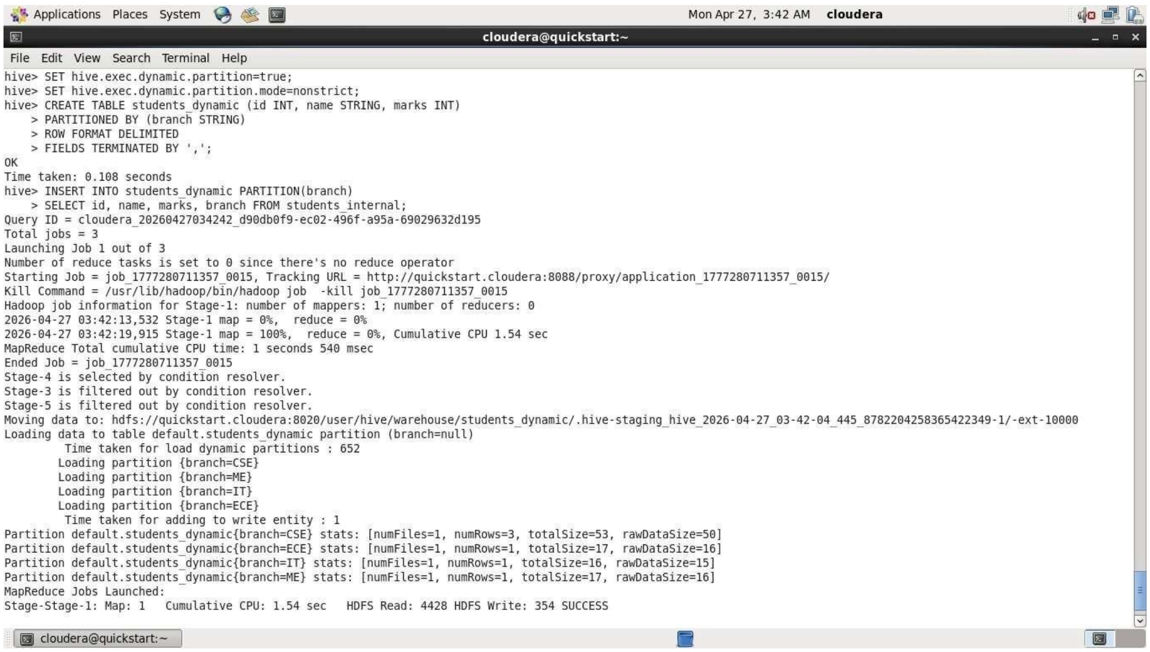


*Screenshot 3: Static partition table created. CSE partition inserted. SELECT * shows 3 CSE students — Aarav, Rohit, Anjali.*

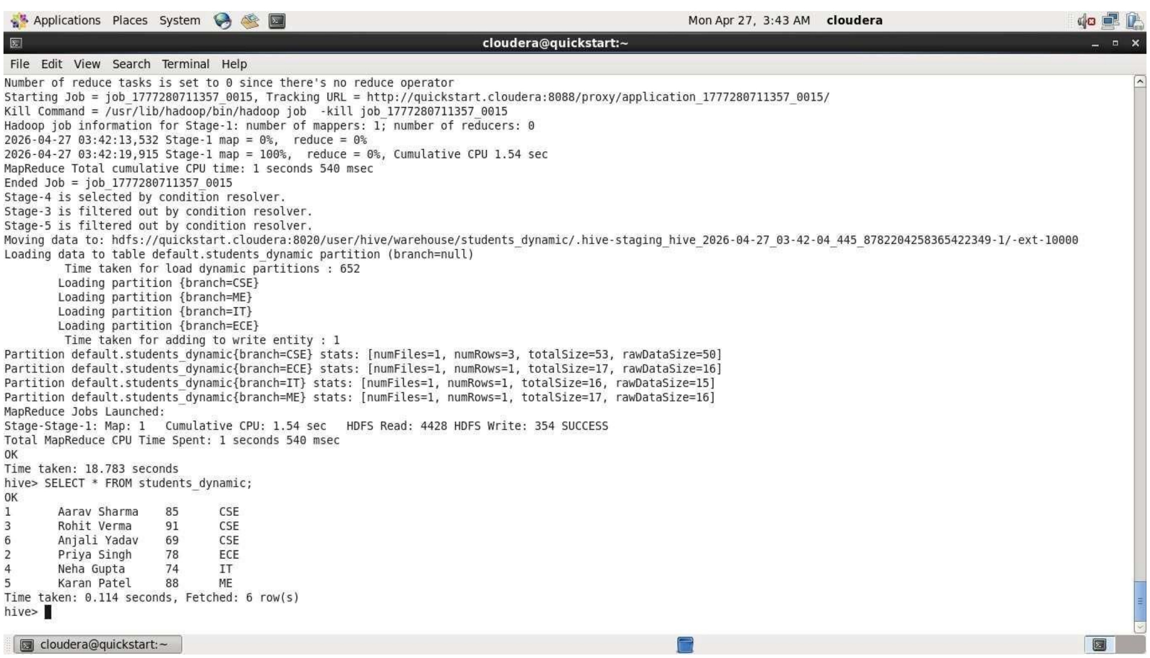
4. Dynamic Partition — students_dynamic

Commands: SET hive.exec.dynamic.partition=true; SET hive.exec.dynamic.partition.mode=nonstrict; Then CREATE TABLE students_dynamic (same structure) PARTITIONED BY (branch STRING). Then INSERT INTO students_dynamic PARTITION(branch) SELECT id, name, marks, branch FROM students_internal;

Dynamic partitioning figures out the partition value automatically from the data. Hive created 4 partitions (CSE, ECE, IT, ME) all at once from the single INSERT. Much easier than static when you have many partition values.



Screenshot 4: Dynamic partitions loading — CSE, ME, IT, ECE all created automatically. Each partition shows correct row counts.



*Screenshot 5: SELECT * FROM students_dynamic — all 6 students returned, sorted by branch partition.*